AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 36 · STUDENT LESSON

ML Builder Expo

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Weeks 15–17: recover metrics from a complete confusion matrix.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Defend the project using traceable evidence. Try the recall before opening notes.

Your learning target

I can defend the project using traceable **evidence**.

I can explain errors and limits to a non-specialist audience.

I can respond to critique and propose an **evidence**-based next step.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone Train, Analyze, Improve. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A builder defends the complete **evidence** chain: problem, data, code, model, metrics, errors, fairness, **limitation**s, and human oversight.



Your target

LEARNING TARGET Final counts are TP=12,FN=3,FP=5,TN=20. Compute accuracy, precision and recall, then state one **limitation** and a review trigger for the model **defense**.

Builder name and experiment date:

Readiness for the connected explanation

What is the difference between a result and a claim?

This week's end goal

By the end, I can explain ml builder expo using a traced example and a stated **limitation**.

STUDENT LESSON · 2 OF 18

Notice the evidence

Investigate the supplied ML Builder Expo case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use ml builder expo to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

The final **defense** tells a coherent story: question, data, method, comparison, errors and limits. A strong project may conclude that the baseline is preferable or **evidence** is insufficient.

The aim is sound reasoning and a reproducible investigation, not an impressive headline or polished unsupported claim.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines machine learning project **defense**.

STUDENT LESSON · 3 OF 18

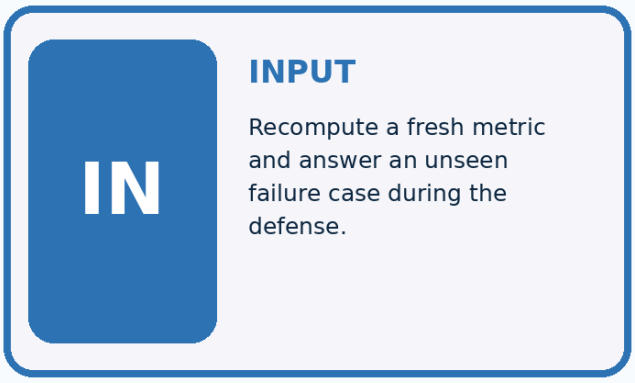
See how it works

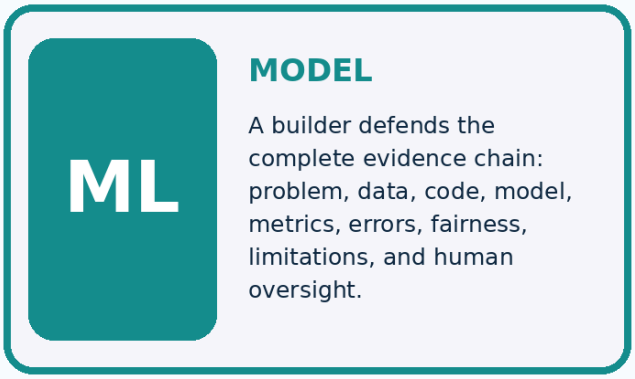
> Follow the mechanism for ML Builder Expo. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Recompute a fresh metric and answer an unseen failure case during the

defense.

MODEL IDEA A builder defends the complete **evidence** chain: problem, data, code, model, metrics, errors, fairness, **limitation**s, and human oversight.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated **limitation**.

How the mechanism works

UNDER THE HOOD Trace one leaf from source and split through pre**process**ing, model, **prediction**, and review. Recompute precision or recall from a fresh confusion matrix without notes.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Example opening: We compared a simple classifier with an always-clear **training** baseline on fictional parcels. We froze the procedure before a 12-case test. Candidate accuracy was 9/12 versus baseline 8/12; candidate FP 1 FN 2.

These small synthetic results support a classroom comparison, not real parcel decisions. We would next investigate dim-condition coverage using new development data.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

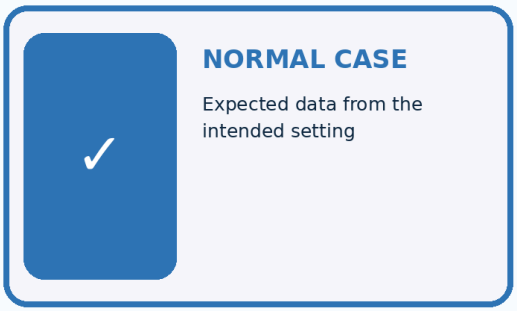
Compare the cases

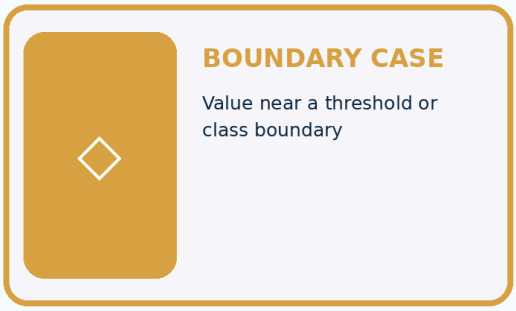
> Compare the supplied cases for ML Builder Expo. Identify the detail that changes the conclusion.

EVALUATION DESIGN

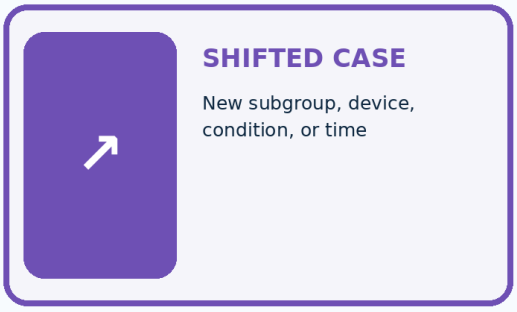
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: The poster omits the baseline and calls the model “field ready”.

Predicted result

Prediction: The headline exceeds the supplied **evidence**.

Why the result follows

Reason: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of ML Builder Expo to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
TECHNICAL DEFENSE
evidence-based explanation under questions
2
TRACEABILITY
link claim to data, code, result, and decision
3
TRANSFER TEST
fresh case not rehearsed
4
RELEASE DECISION
approve, limit, revise, or stop

Term	Precise meaning	My ML example
technical defense	evidence-based explanation under questions	
traceability	link claim to data, code, result, and decision	
transfer test	fresh case not rehearsed	
release decision	approve, limit, revise, or stop	

> MISCONCEPTION TO REPAIR A smooth prepared demo proves the model is reliable and ready for use.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Defense	An explanation of decisions supported by inspectable evidence
Traceability	Ability to follow a claim back to its source
Limitation	A boundary on interpretation or supported use
Reflection	An account of learning and justified next steps

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for ML Builder Expo, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 36 • WORKED EVIDENCE

1. Trace one leaf from source and split through preprocessing, model, prediction, and review.
2. Recompute precision or recall from a fresh confusion matrix without notes.
3. Run an unseen low-light or missing-feature case and follow the documented fallback.
4. State intended use, do-not-use, subgroup limitation, provenance, and who remains accountable.

Step 1

STEP 1 Trace one leaf from source and split through pre**process**ing, model, **prediction**, and review.

Step 2

STEP 2 Recompute precision or recall from a fresh confusion matrix without notes.

Step 3

STEP 3 Run an unseen low-light or missing-feature case and follow the documented fallback.

Step 4

STEP 4 State intended use, do-not-use, subgroup **limitation**, provenance, and who remains accountable.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Machine learning project defense at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Show the process	Task data splits versions	Explain one prediction
Show the evidence	Baseline metrics errors	Trace one reported number
Show the limits	Unsupported uses next step	Answer peer questions

Prepare a five-minute presentation with a one-page evidence summary and linked project files. Include the data dictionary, split record, run record, prediction ledger, model card and revision history. Demonstrate one prediction or arithmetic calculation live so the audience can follow the reasoning.

Name the files needed to audit one reported accuracy figure. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: ML Builder Expo

Present a plant-classifier expo claim that another team could check.

The facts we will use

Supplied facts

Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct. Three candidate failures are recorded; no outdoor trial exists.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/36

ML Builder Expo

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Show denominator

Candidate12/15=80%; baseline10/15≈66.7% on the same cases.



2. Show independence

Explain five held-out plant identities and three related views per plant.



3. Show failures

Present the three failure records and indoor-only limitation beside the headline rate.

CHANGE ONE THING

The poster omits the baseline and calls the model "field ready".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Show denominator	Candidate12/15=80%; baseline10/15≈66.7% on the same cases.
2. Show independence	Explain five held-out plant identities and three related views per plant.
3. Show failures	Present the three failure records and indoor-only limitation beside the headline rate.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Clear presentation helps review but does not increase the strength of a tiny or unrepresentative evaluation.

Keep the stated limit

Read the labelled picture: Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The poster omits the baseline and calls the model “field ready”. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the ML Builder Expo claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A smooth prepared demo proves the model is reliable and ready for use. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A smooth prepared demo proves the model is reliable and ready for use.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

The teacher rubric has four dimensions scored 0, 1, 2: valid data/splits, reproducible method, complete evaluation, and honest communication.

Two means **evidence** is complete and explainable; one means partial **evidence** with identified gaps; zero means absent or materially incorrect.

Report dimensions separately; a serious leakage problem cannot be hidden by presentation points.

Write one next investigation tied to an observed limitation, and identify what must stay held out.

STUDENT LESSON · 8 OF 18

Practise together

> Practise ML Builder Expo with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + **limitation**s

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Peers ask three questions: what influenced model selection, which cases failed, and what **evidence** would change the conclusion? Respond using saved records. If an answer is unknown, say so and name the **evidence** needed.

Record feedback as resolved, deferred with reason, or requiring new data.

Practise with less help: Trace evidence

Use workbook W2 for the connected example “Machine learning project **defense**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Machine learning project **defense**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

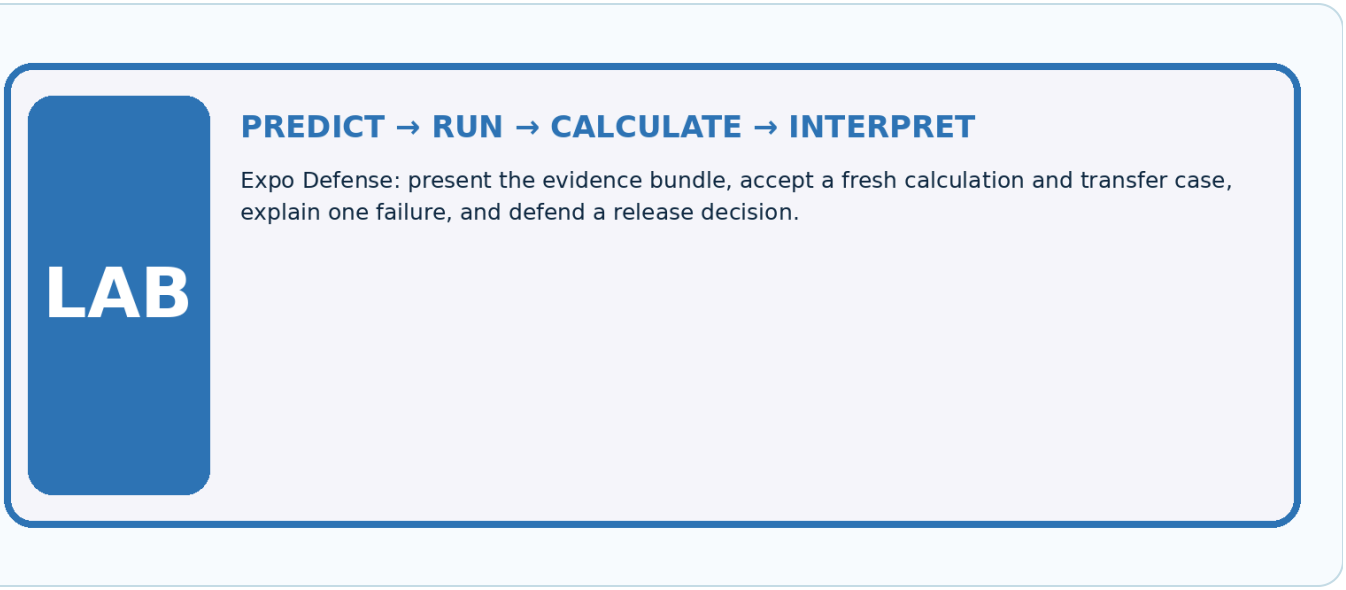
Run the investigation

> Investigate ML Builder Expo. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Expo **Defense**: present the **evidence** bundle, accept a fresh calculation and transfer case, explain one failure, and defend a **release decision**.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
> **Prediction:** _____
> **Actual:** _____

> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

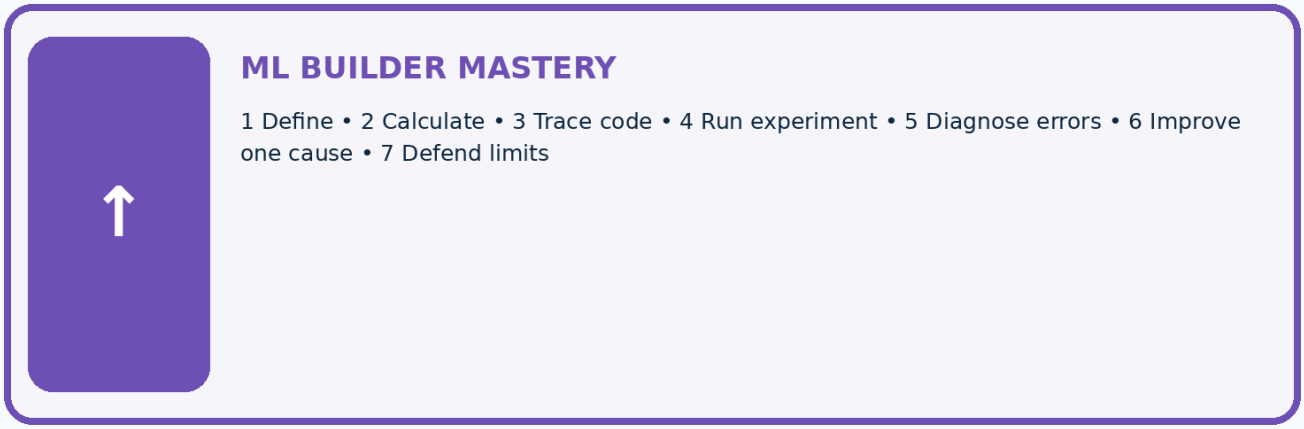
Build your understanding

> Move from recall to tracing, diagnosis and transfer in ML Builder Expo. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **technical defense**, **traceability**, **transfer test**, **release decision**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

> TRACE Explain the data, code or decision path.

> DIAGNOSE A smooth prepared demo proves the model is reliable and ready for use.

> TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

> DEFEND State intended use, metric, **limitation**, and human responsibility.

> MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Deliver the final Responsible Plant Classifier product card, live **evidence** trace, model card, and human-oversight commitment.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

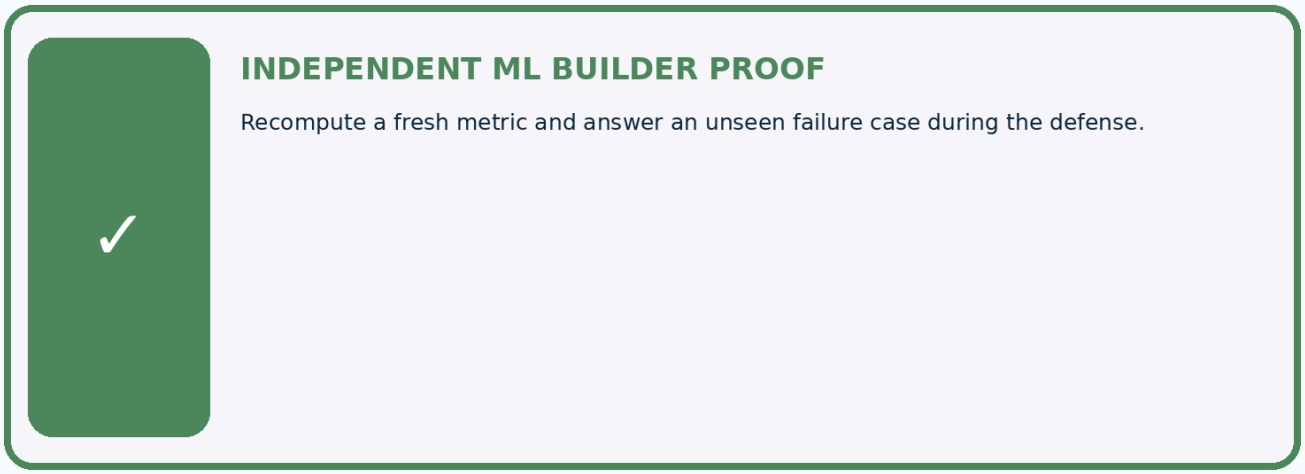
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Recompute a fresh metric and answer an unseen failure case during the **defense**.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a **limitation**, fairness concern, and the human-review action.

Independent **evidence** and **defense**:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: AI-5 AI Programmer.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete ML Builder Expo case for supported practice. Repeated examples do not become fresh assessments.

Week 36 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Final classifier gets16/20 correct; two of four errors are false negatives.

The question

Report supported metrics and what is missing.

Worked explanation

Accuracy**80%**. Recall cannot be computed from FN alone; actual positives or TP are needed. Show full matrix, limits and review path.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Report only metrics the evidence supports

Evidence or stage	Value or result
Correct final predictions	16/20

Final classifier gets16/20 correct; two of four errors are false negatives.

Report supported metrics and what is missing.

Accuracy**80%**. Recall cannot be computed from FN alone; actual positives or TP are needed. Show full matrix, limits and review path.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Review a **defense** claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 36 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Your expo model succeeds on three demonstration plants. What **evidence** should accompany the demo and what **limitation** should you state?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Deliver the final Responsible Plant Classifier product card, live **evidence** trace, model card, and human-oversight commitment.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Final counts are TP=12,FN=3,FP=5,TN=20. Compute accuracy, precision and recall, then state one **limitation** and a review trigger for the model **defense**.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate ML Builder Expo, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Present a plant-classifier expo claim that another team could check.

By the end, I can	How I will show it
Defend the project using traceable evidence	Explain the deciding step, show a trace or test, and state one limit.
Explain errors and limits to a non-specialist audience	Explain the deciding step, show a trace or test, and state one limit.
Respond to critique and propose an evidence-based next step	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain ML Builder Expo without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Defend the project using traceable **evidence**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Present a plant-classifier expo claim that another team could check.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Present the three failure records and indoor-only **limitation** beside the headline rate.

1. Show failures

2. Show independence

3. Show denominator

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Show failures Present the three failure records and indoor-only **limitation** beside the headline rate. Read the picture in order.

Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The poster omits the baseline and calls the model “field ready”. Which conclusion is supported by the supplied facts?

1. The headline exceeds the supplied **evidence**.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The headline exceeds the supplied **evidence**. No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Repair a missing building block

Review Capstone Train, Analyze, Improve (ai-4/35). Goal: Run a frozen final evaluation with complete accounting. Short practice: Why is “the revision is better” incomplete?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Cards and Human Review (ai-4/33). Goal: Write a model card whose claims match **evidence**. Short practice: Write a bounded replacement for the publicity sentence.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Improve Generalization (ai-4/30). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: What fresh **evidence** is needed after adding crossed **training** data?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, **reflection**s, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

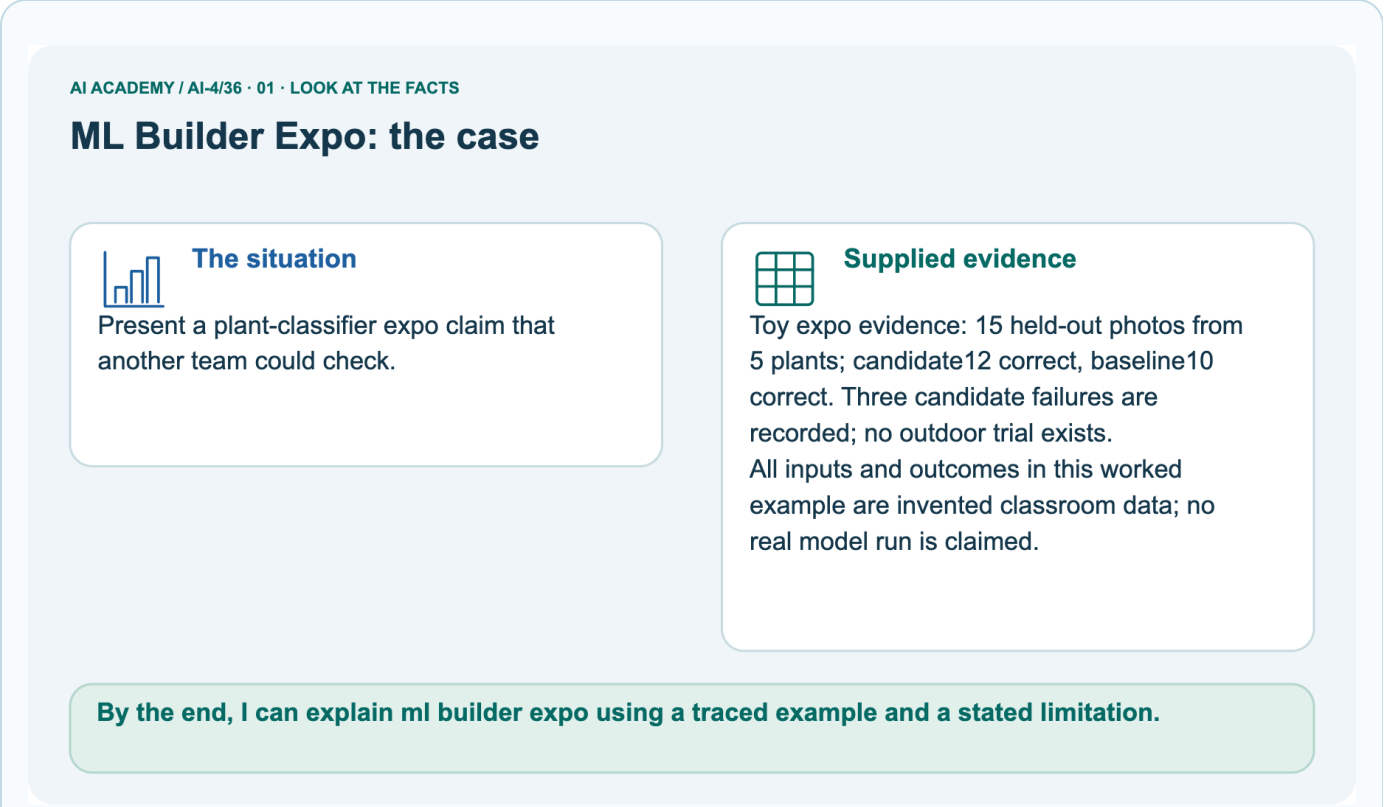
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each ML Builder Expo graphic using its labels, deciding evidence and relevant limitation.

01. ML Builder Expo: the case



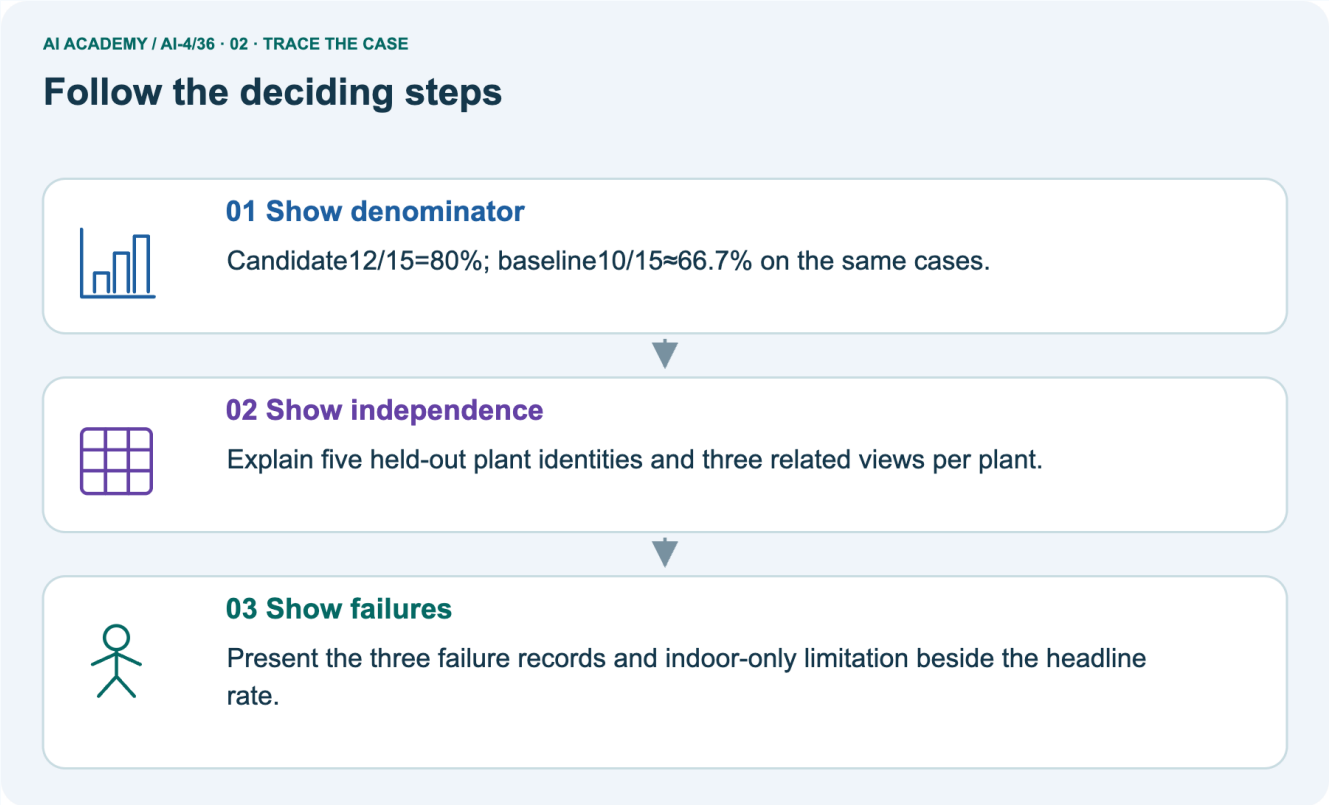
Follow the labelled picture with the matching explanation. Open the vector diagram

Present a plant-classifier expo claim that another team could check. Supplied facts: Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct.

Three candidate failures are recorded; no outdoor trial exists. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

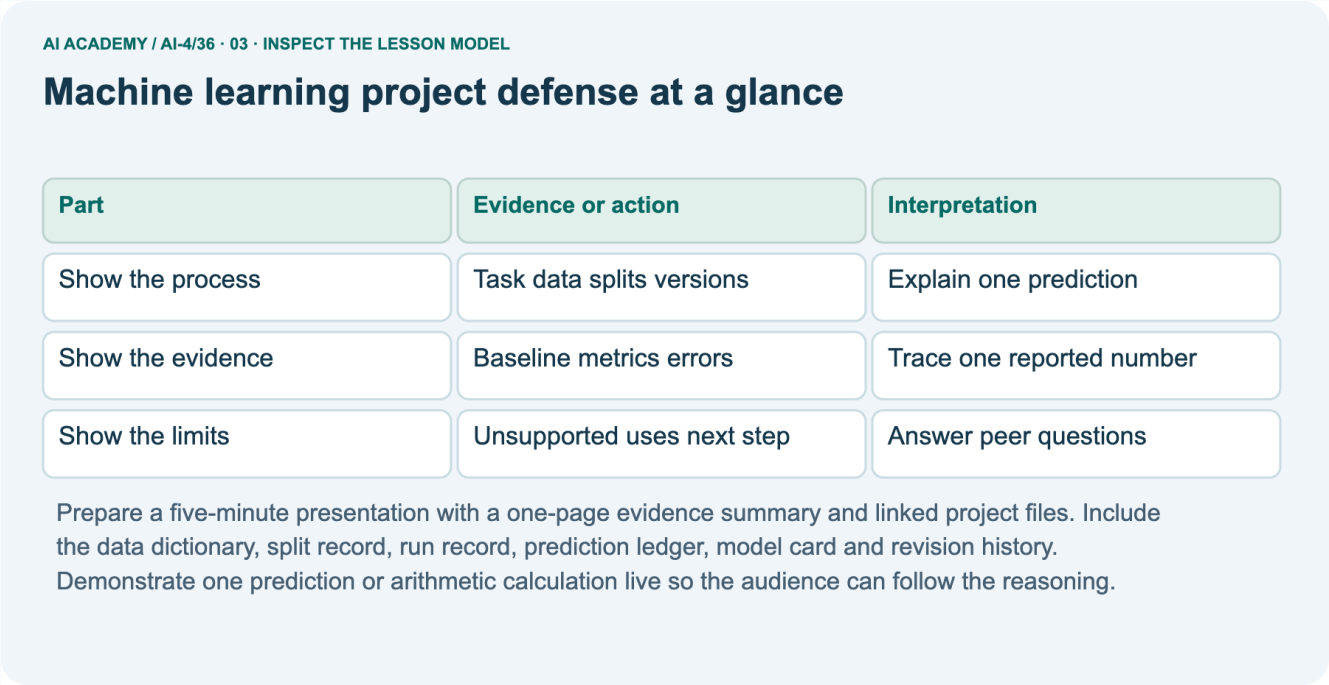


Follow the labelled picture with the matching explanation. Open the vector diagram

Show denominator: Candidate12/15=**80%**; baseline10/15≈66.7% on the same cases. → Show independence: Explain five held-out plant identities and three related views per plant. → Show failures: Present the three failure records and indoor-only **limitation** beside the headline rate.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Machine learning project defense at a glance



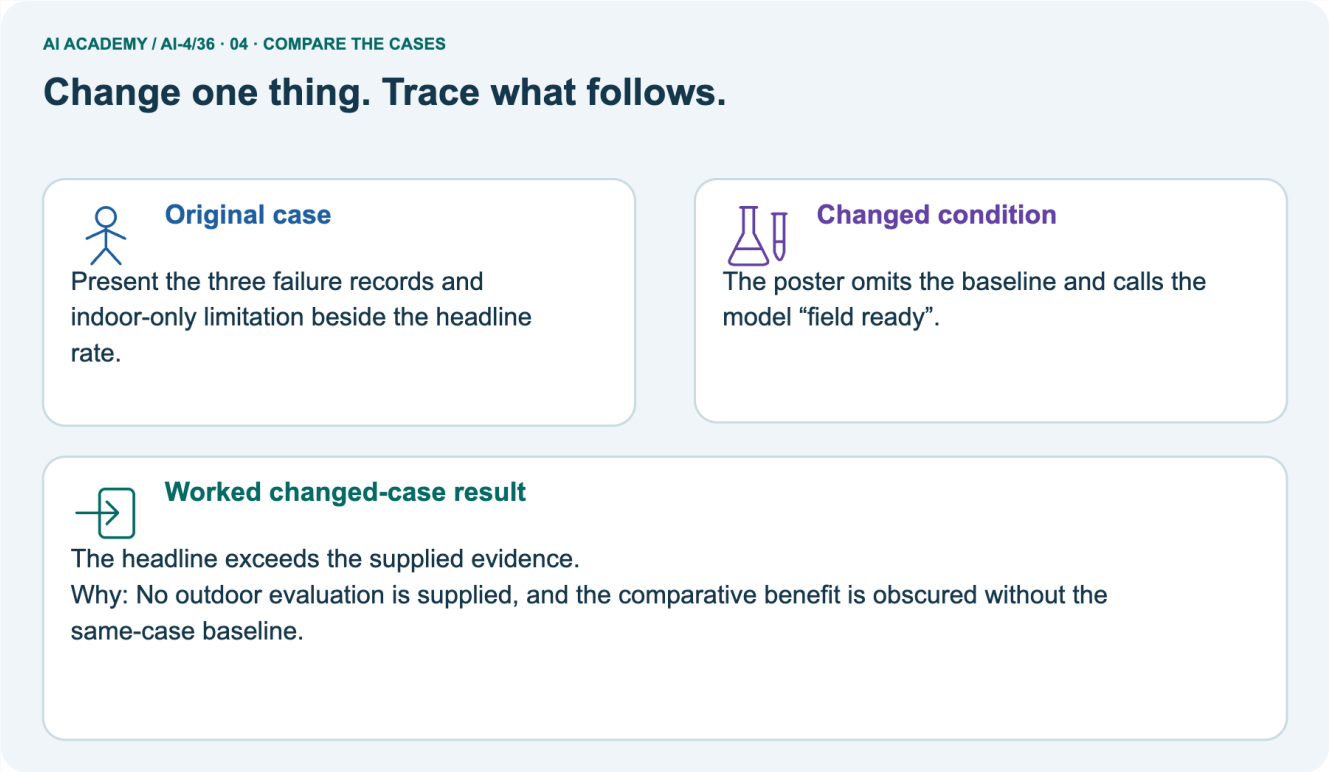
Follow the labelled picture with the matching explanation. Open the vector diagram

Machine learning project **defense** at a glance. Show the **process** | Task data splits versions | Explain one **prediction**. Show the **evidence** | Baseline metrics errors | Trace one reported number.

Show the limits | Unsupported uses next step | Answer peer questions

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



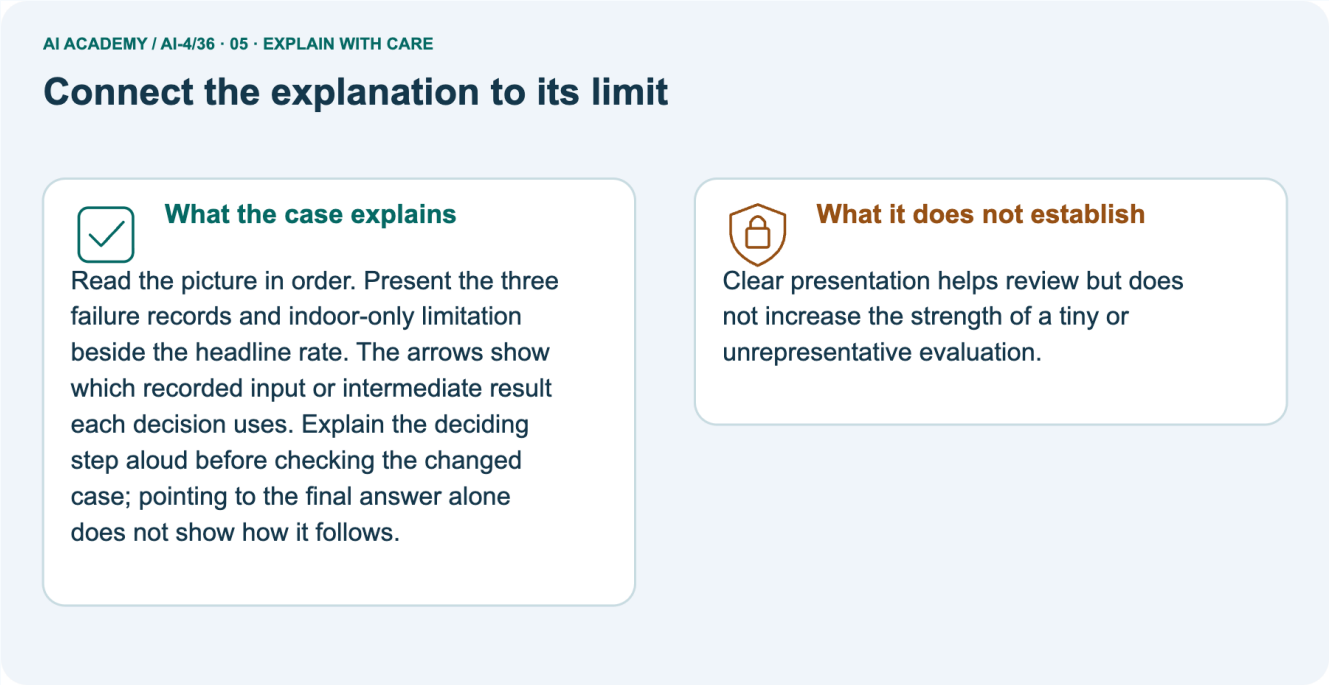
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Present the three failure records and indoor-only **limitation** beside the headline rate. Changed condition: The poster omits the baseline and calls the model “field ready”.

Expected result: The headline exceeds the supplied **evidence**. Reason: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

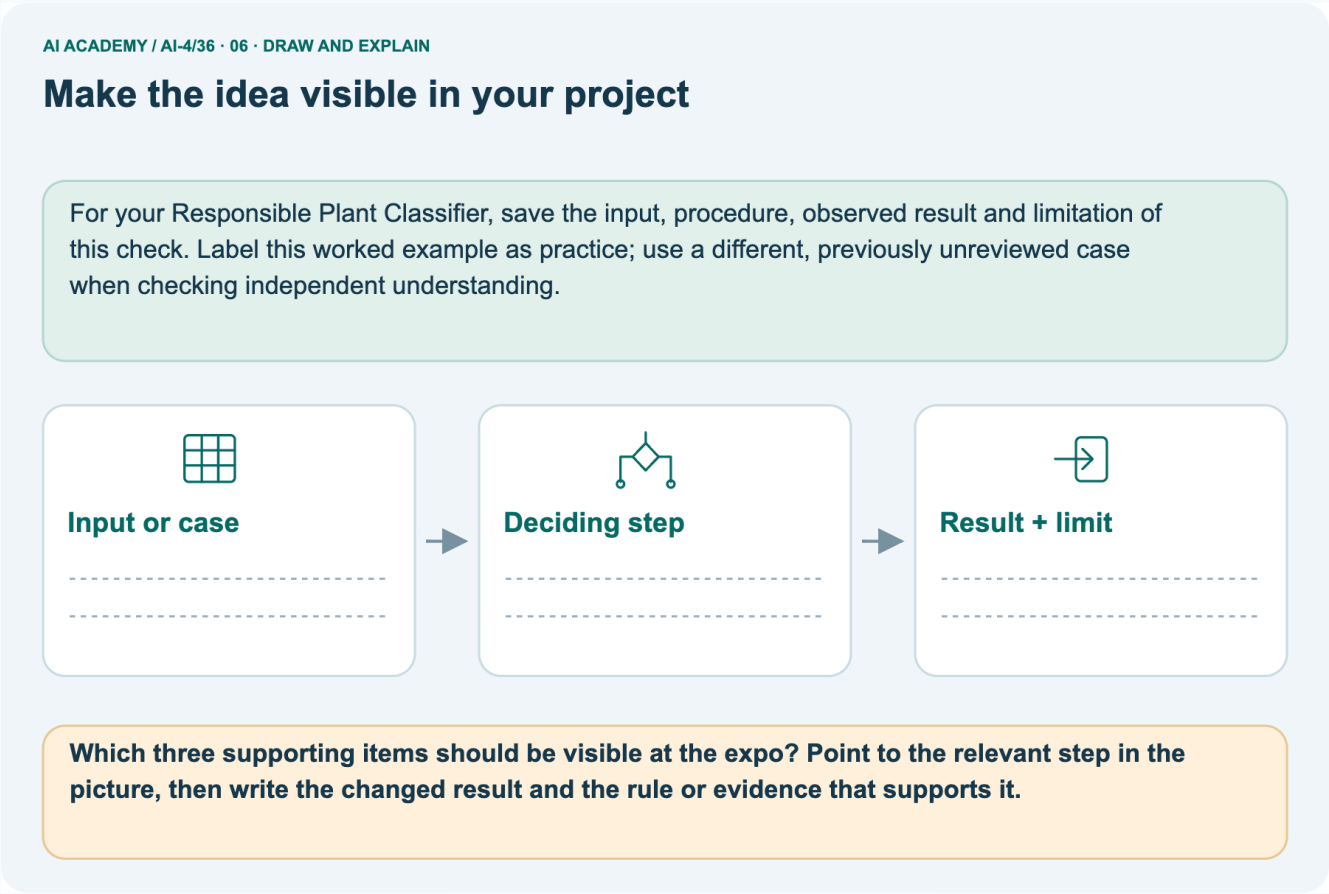
Read the picture in order. Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Clear presentation helps review but does not increase the strength of a tiny or unrepresentative evaluation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.